

Lab 1: Packet analysis at application layer using Wireshark

SCSR1213 Network Communications

Universiti Teknologi Malaysia

Objective:

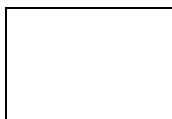
1. Understanding of network protocols by observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences.
2. To introduce student with Wireshark software tool for packet analyzer.
3. To analyze protocol used in application layer such as http and dns.

Reference material: Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross.

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Section :02



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PART A: Wireshark Getting Started

1.0 Introduction

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures (“sniffs”) messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure A.1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure A.1 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. In Figure A.1, the assumed physical media is an Ethernet, and so all upper-layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

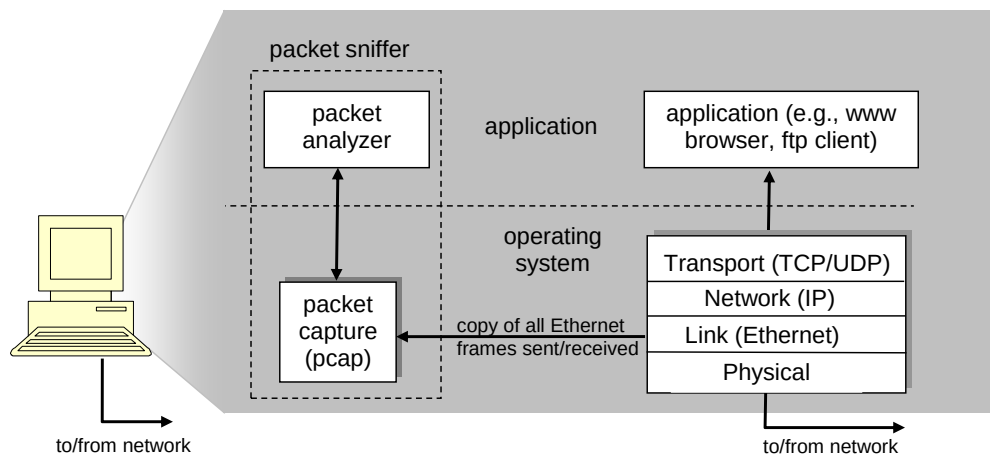


Figure A.1: Packet sniffer structure

The second component of a packet sniffer is the **packet analyzer**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must “understand” the structure of all messages exchanged by protocols. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string “GET,” “POST,” or “HEAD”.

2.0 Getting Wireshark Ready

- Download and install the Wireshark software
- Run Wireshark. Wireshark startup screen shown in Figure A.2.

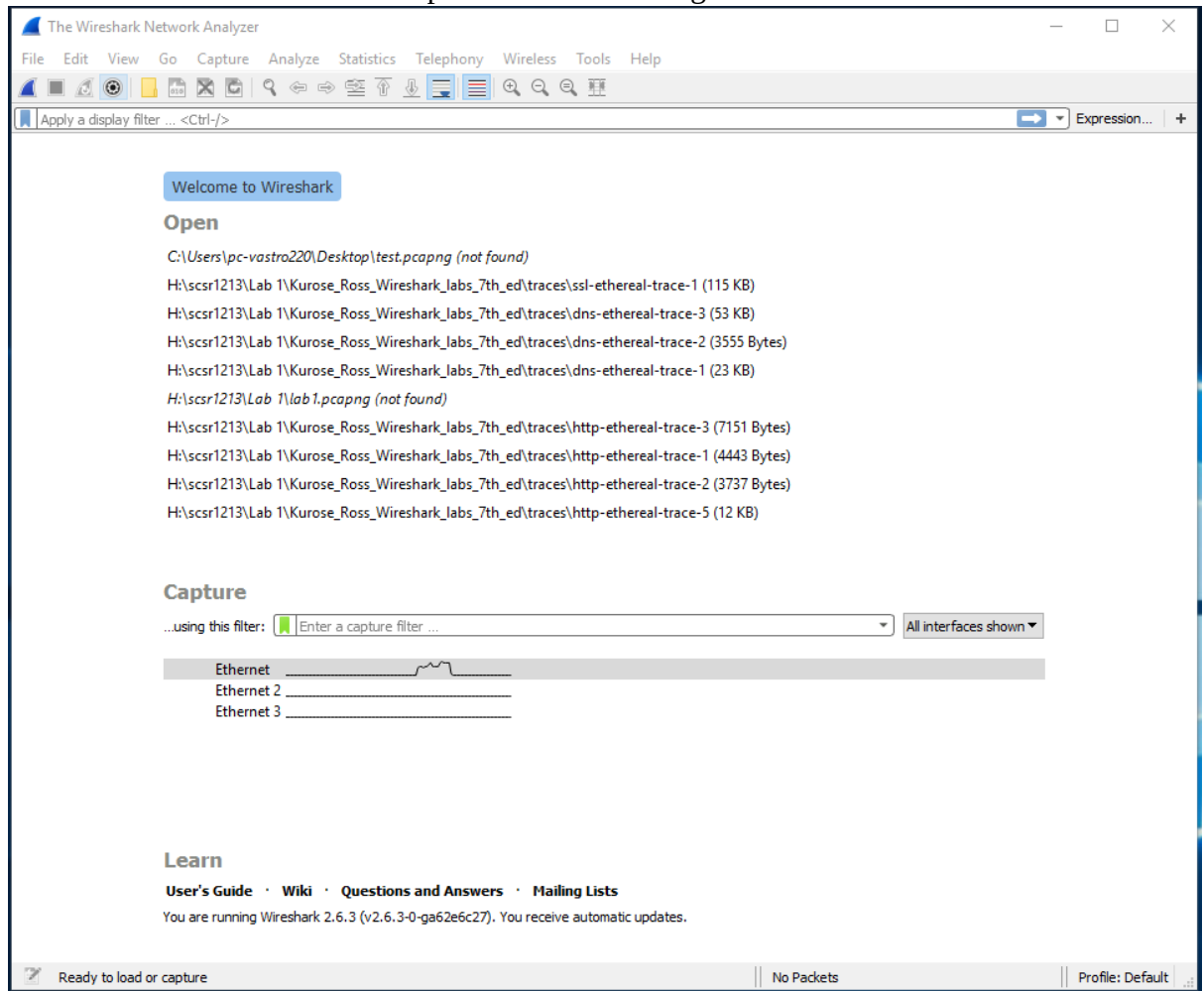


Figure A.2: Initial Wireshark startup screen

- The Wireshark interface has five major components as shown in Figure A.3.

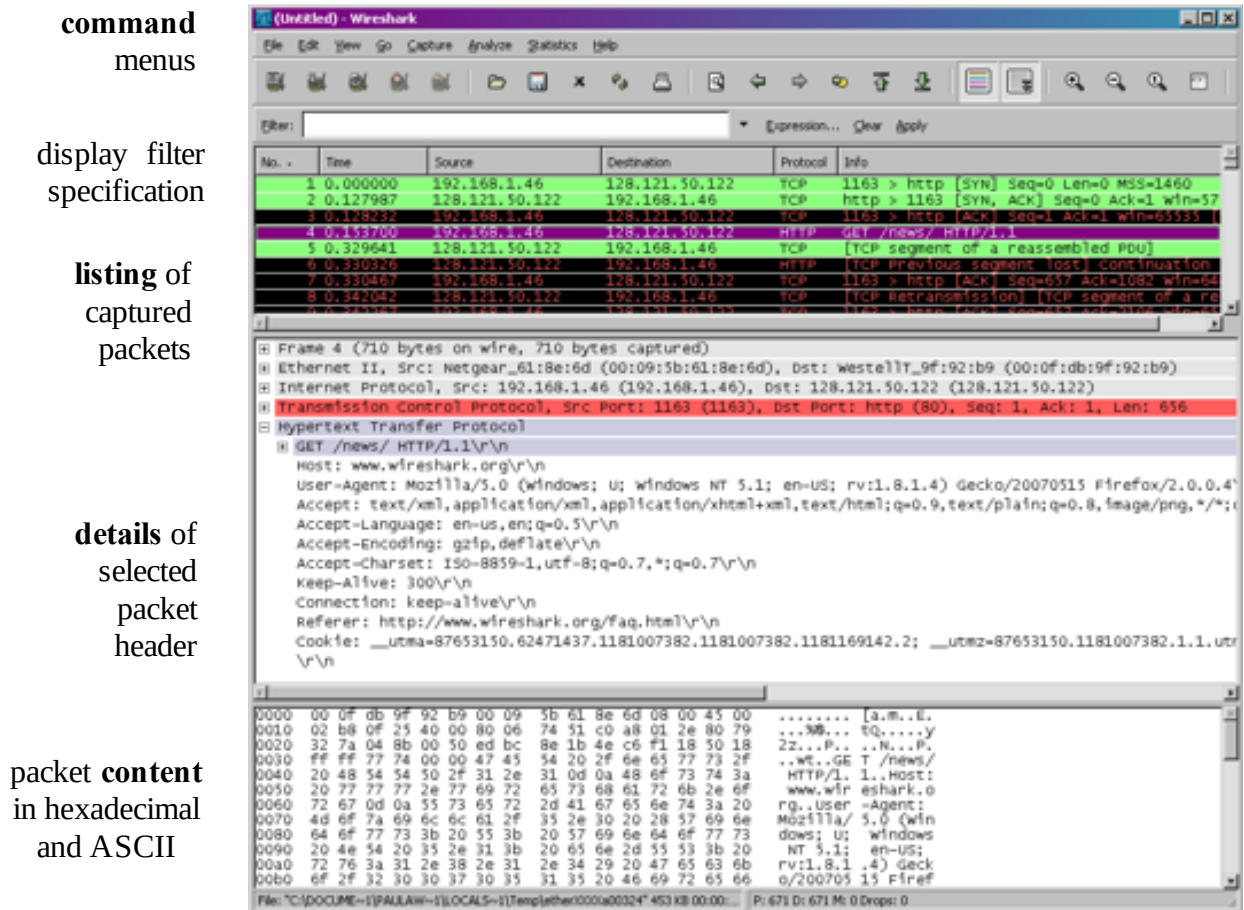


Figure A.3: Wireshark Graphical User Interface, during packet capture and

- The **command menus** are standard pulldown menus located at the top of the window.
- The **packet display filter field**, into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number, the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. These details include information about the Ethernet frame and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.
- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

3.0 Test Run Wireshark

- Start up the Wireshark software.
- To begin packet capture, select the Capture pull down menu and pick Options menu. Select appropriate interfaces on your compute and click Start button to begin packet capture. Refer to Figure A.4

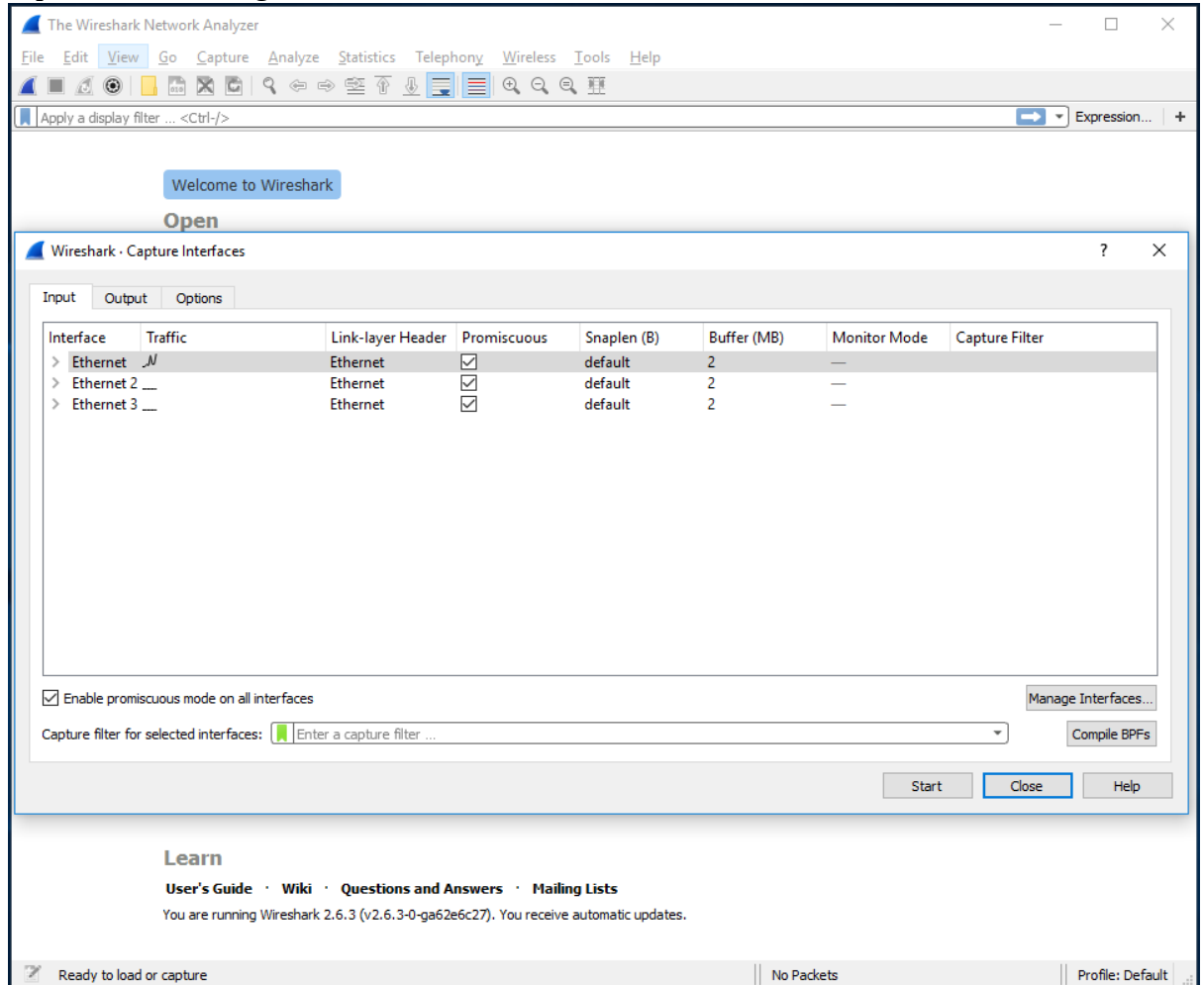
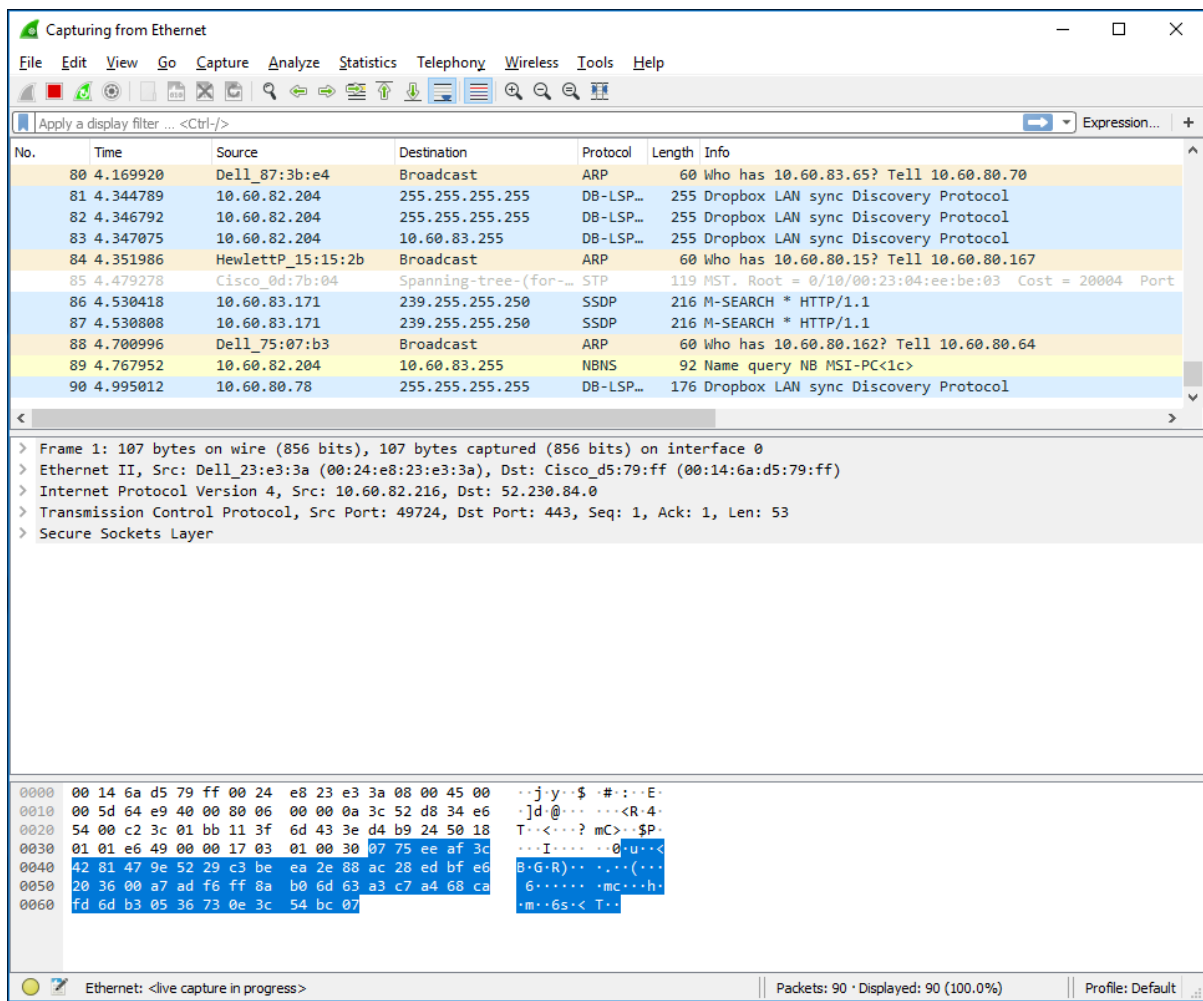


Figure A.4: Capture and Options Menu

- Once you begin packet capture, result will be shown as in Figure A.5.



The image shows the Wireshark interface with a packet capture in progress. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons. A display filter bar shows 'Apply a display filter ... <Ctrl-/>'. The main packet list table is as follows:

No.	Time	Source	Destination	Protocol	Length	Info
80	4.169920	Dell_87:3b:e4	Broadcast	ARP	60	Who has 10.60.83.65? Tell 10.60.80.70
81	4.344789	10.60.82.204	255.255.255.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
82	4.346792	10.60.82.204	255.255.255.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
83	4.347075	10.60.82.204	10.60.83.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
84	4.351986	HewlettP_15:15:2b	Broadcast	ARP	60	Who has 10.60.80.15? Tell 10.60.80.167
85	4.479278	Cisco_0d:7b:04	Spanning-tree-(for-...	STP	119	MST. Root = 0/10/00:23:04:ee:be:03 Cost = 20004 Port
86	4.530418	10.60.83.171	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
87	4.530808	10.60.83.171	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
88	4.700996	Dell_75:07:b3	Broadcast	ARP	60	Who has 10.60.80.162? Tell 10.60.80.64
89	4.767952	10.60.82.204	10.60.83.255	NBNS	92	Name query NB MSI-PC<1c>
90	4.995012	10.60.80.78	255.255.255.255	DB-LSP...	176	Dropbox LAN sync Discovery Protocol

Below the packet list, the details pane shows the selected packet (Frame 1) with the following information:

- > Frame 1: 107 bytes on wire (856 bits), 107 bytes captured (856 bits) on interface 0
- > Ethernet II, Src: Dell_23:e3:3a (00:24:e8:23:e3:3a), Dst: Cisco_d5:79:ff (00:14:6a:d5:79:ff)
- > Internet Protocol Version 4, Src: 10.60.82.216, Dst: 52.230.84.0
- > Transmission Control Protocol, Src Port: 49724, Dst Port: 443, Seq: 1, Ack: 1, Len: 53
- > Secure Sockets Layer

The bottom pane shows the raw packet data in hexadecimal and ASCII format.

Figure A.5: Wireshark packet capture result

- By selecting Capture pulldown menu and selecting Stop, you can stop packet capture.

- Type “arp” in packet display filter field and press Enter key. This will cause only ARP message to be displayed in the packet-listing window as shown in Figure A.6.

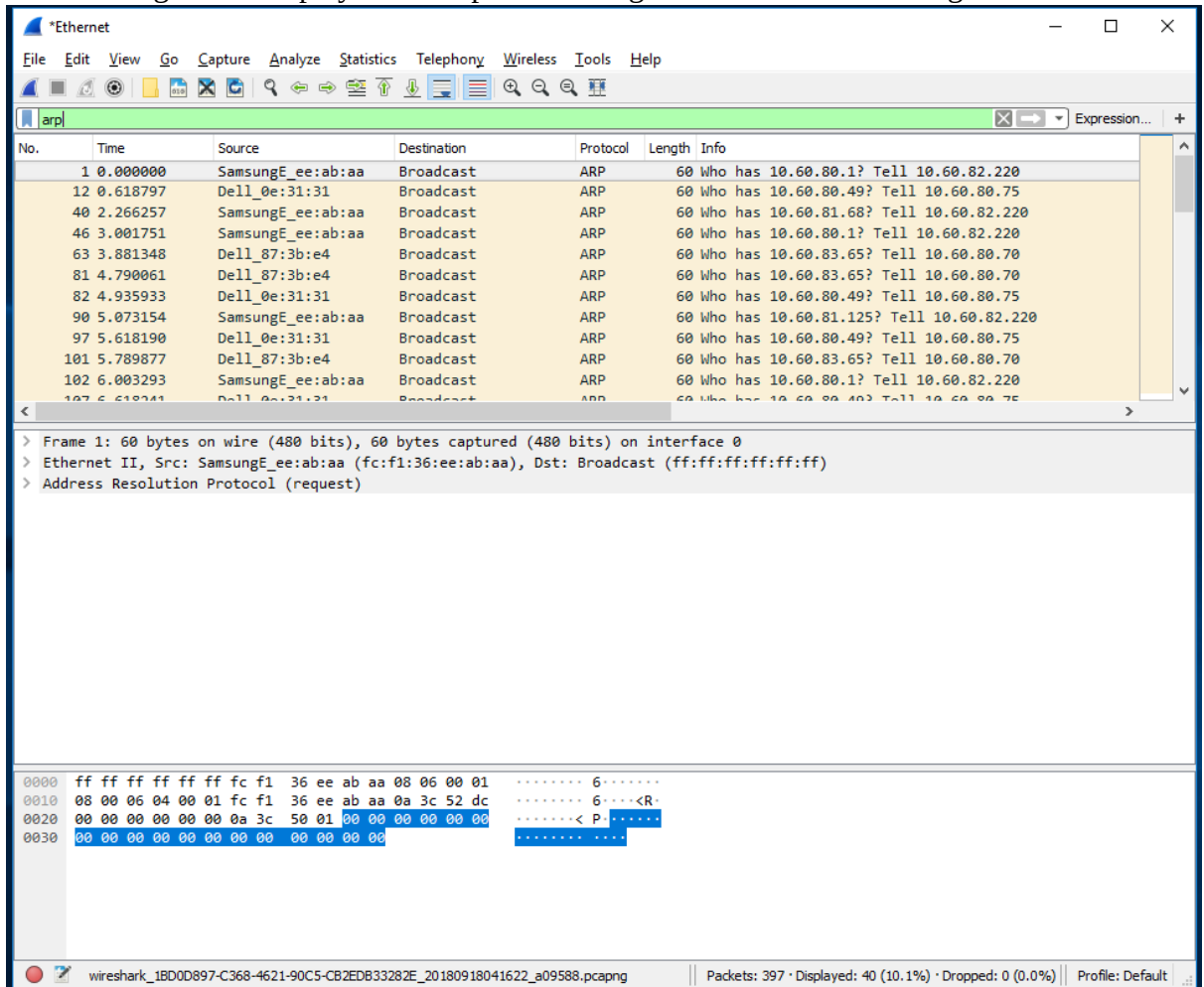


Figure A.6: ARP packet capture

- To save the trace result, use File pulldown menu and select Save function as shown in Figure A.7.

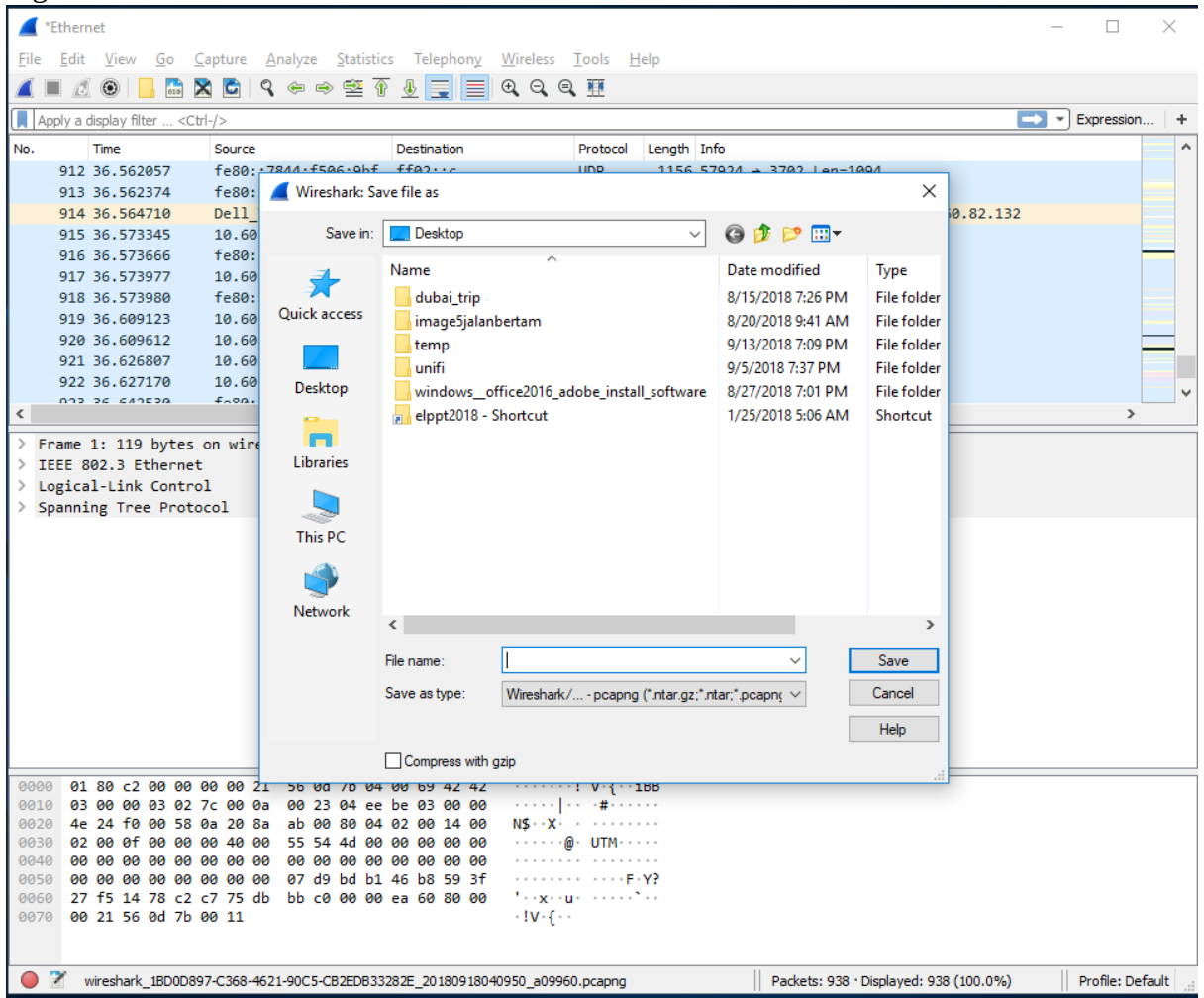


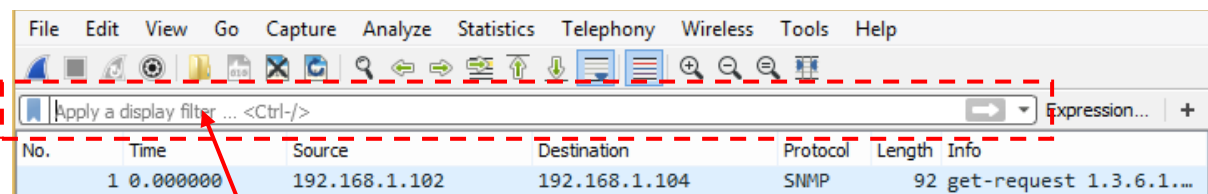
Figure A.7: Save Wireshark trace result

PART B: HTTP Trace

In this part, we'll explore several aspects of the HTTP protocol: the basic GET/response interaction, HTTP message formats and retrieving HTML files with embedded objects. Before beginning these labs, you might want to review Section 2.2 of the textbook.

B.1 The Basic HTTP GET/response interaction

- Open packet trace file **lab1-http-B01.pcapng**.
- Enter “**http**” (just the letters, not the quotation marks) in the **packet display filter field**, so that only captured HTTP messages will be displayed later in the packet-listing window. Refer to figure below:



packet display filter

- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. What version of HTTP is the server running?
HTTP/1.1
2. What is the IP address of the client computer?
192.168.1.102
3. What is the IP address of the gaia.cs.umass.edu server?
128.119.245.12
4. How many bytes of content are being returned to client browser?
439 + 1395 = 1834 bytes
5. What is the status code returned from the server to client browser?
200 OK and 404 Not Found

B.2 The HTTP CONDITIONAL GET/response interaction

- Open packet trace file **lab1-http-B02.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?

No

2. Inspect the contents of the server response after the first GET request from client. Did the server explicitly return the contents of the file? How can you tell?

The server responds with HTTP/1.1 200 OK and the presence of (text/html) shows that the response body contains full contents of the requested html file.

3. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?

If-Modified-since: Tue, 23 Sep 2003 05:35:00 GMT\r\n

4. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

304 Not Modified. The server did not send the full file contents. Since the file hasn't changed, the server just tells the browser to use the cached version instead of downloading it again in order to avoid wasting resources.

B.3 HTML Documents with Embedded Objects

- Open packet trace file **lab1-http-B03.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. How many HTTP GET request messages did client browser send?

3

2. To which Internet addresses were these GET requests sent?

128.119.245.12

165.193.123.218

134.241.6.82

3. any bytes of content are being returned to client browser for the **pearson-logo-footer.gif** image file?

625

4. How many bytes of content are being returned to client browser for the **cover.jpg** image file?

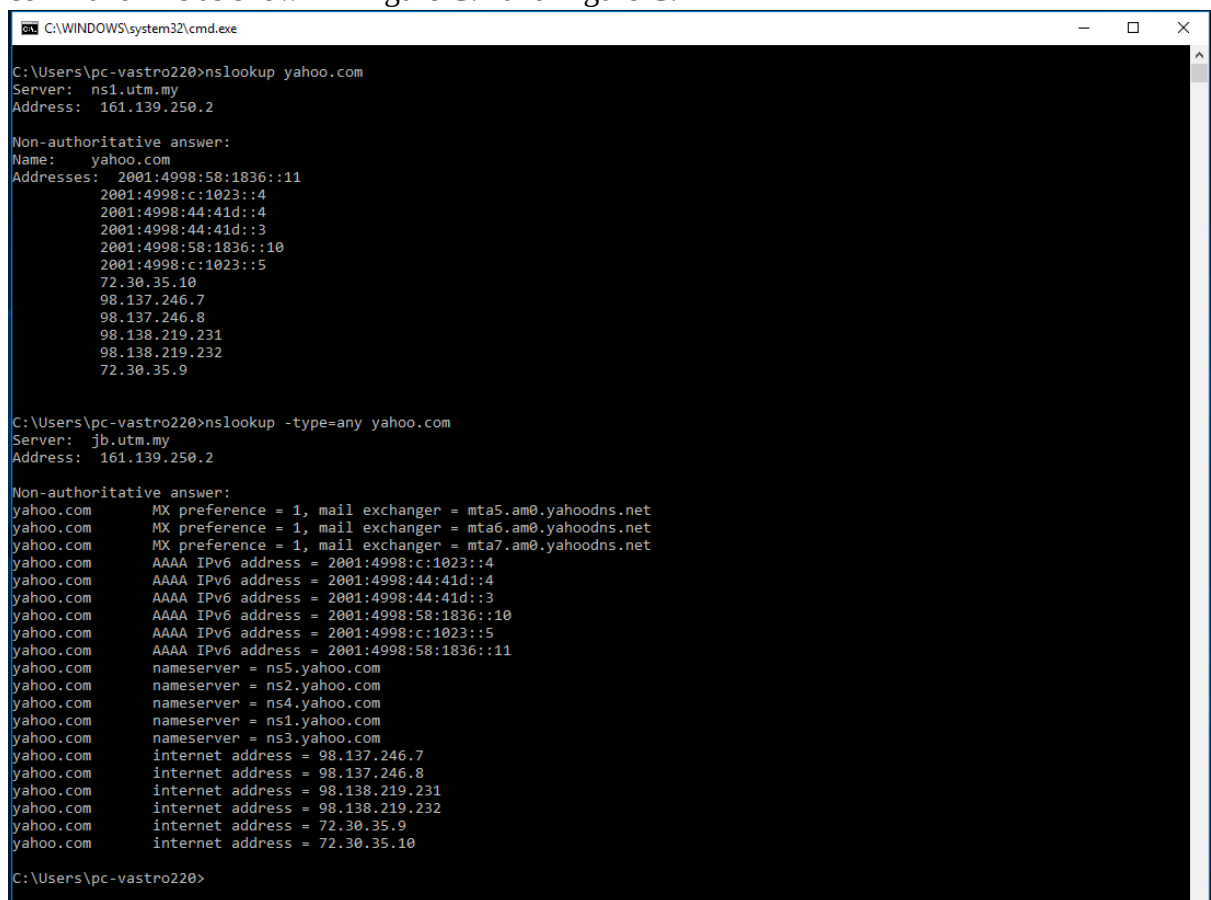
609

PART C: DNS Trace

1.0 nslookup

nslookup tool allows the host running the tool to query any specified DNS server for a DNS record. The queried DNS server can be a root DNS server, a top-level-domain DNS server, an authoritative DNS server, or an intermediate DNS server. To accomplish this task, nslookup sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.

- To run it in Windows, open the Command Prompt (cmd) and run nslookup on the command line as shown in Figure C.1 and Figure C.2



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup yahoo.com
Server: ns1.utm.my
Address: 161.139.250.2

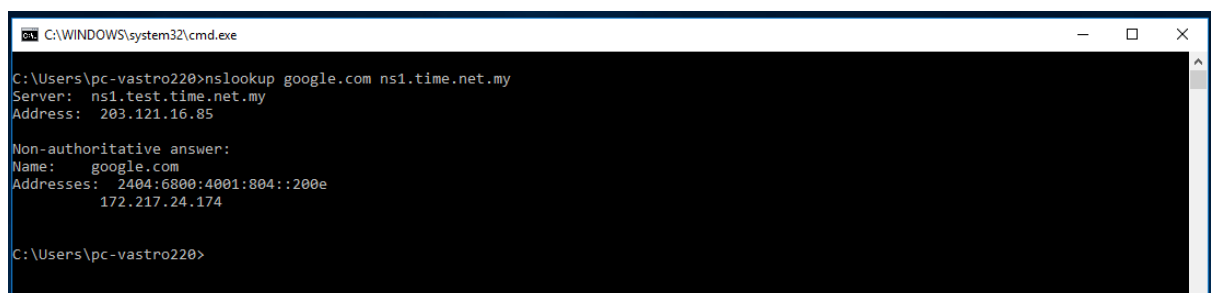
Non-authoritative answer:
Name: yahoo.com
Addresses: 2001:4998:58:1836::11
           2001:4998:c:1023::4
           2001:4998:44:41d::4
           2001:4998:44:41d::3
           2001:4998:58:1836::10
           2001:4998:c:1023::5
           72.30.35.10
           98.137.246.7
           98.137.246.8
           98.138.219.231
           98.138.219.232
           72.30.35.9

C:\Users\pc-vastro220>nslookup -type=any yahoo.com
Server: jlb.utm.my
Address: 161.139.250.2

Non-authoritative answer:
yahoo.com      MX preference = 1, mail exchanger = mta5.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta6.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta7.am0.yahoodns.net
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::3
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::10
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::5
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::11
yahoo.com      nameserver = ns5.yahoo.com
yahoo.com      nameserver = ns2.yahoo.com
yahoo.com      nameserver = ns4.yahoo.com
yahoo.com      nameserver = ns1.yahoo.com
yahoo.com      nameserver = ns3.yahoo.com
yahoo.com      internet address = 98.137.246.7
yahoo.com      internet address = 98.137.246.8
yahoo.com      internet address = 98.138.219.231
yahoo.com      internet address = 98.138.219.232
yahoo.com      internet address = 72.30.35.9
yahoo.com      internet address = 72.30.35.10

C:\Users\pc-vastro220>
```

Figure C.1: nslookup result



```
C:\WINDOWS\system32\cmd.exe

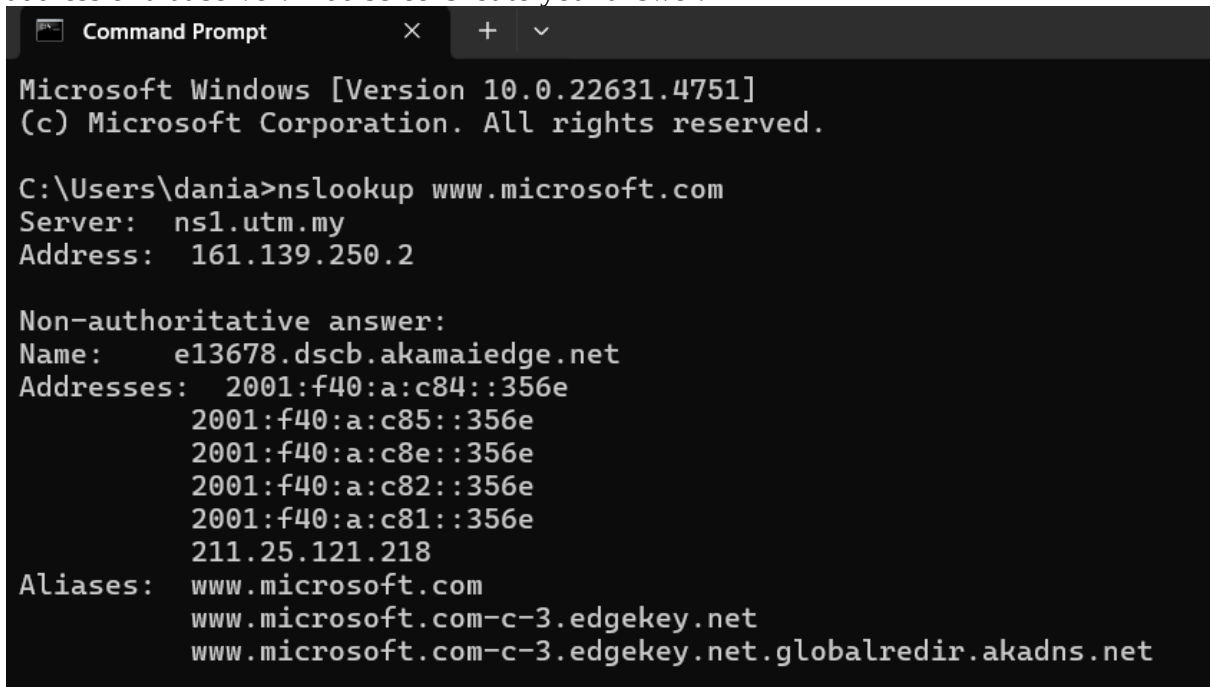
C:\Users\pc-vastro220>nslookup google.com ns1.time.net.my
Server: ns1.test.time.net.my
Address: 203.121.16.85

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4001:804::200e
           172.217.24.174

C:\Users\pc-vastro220>
```

Figure C.2: nslookup result

1. Run nslookup to obtain the IP address of a www.microsoft.com server. What is the IP address of that server? Add screenshot to your answer.

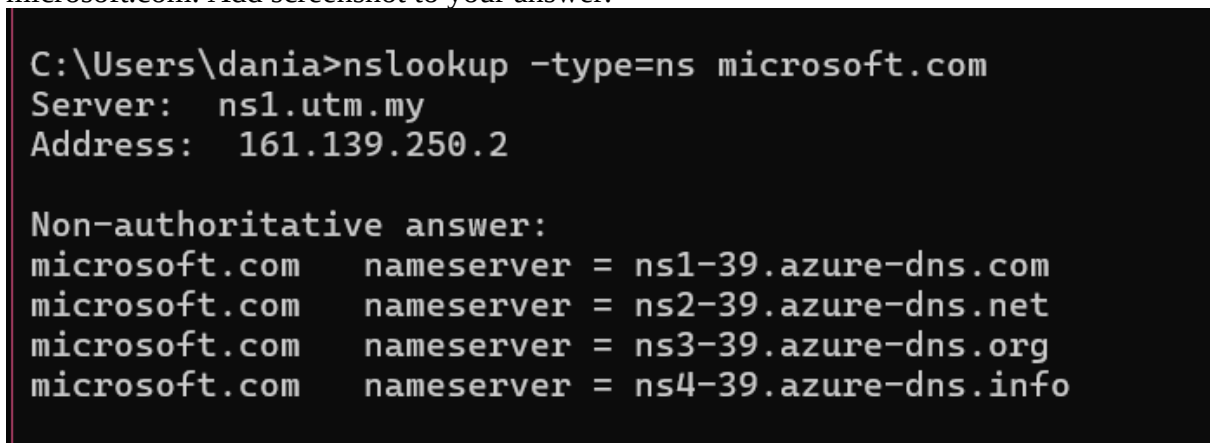


```
Command Prompt
Microsoft Windows [Version 10.0.22631.4751]
(c) Microsoft Corporation. All rights reserved.

C:\Users\дания>nslookup www.microsoft.com
Server:  ns1.utm.my
Address:  161.139.250.2

Non-authoritative answer:
Name:     e13678.dscb.akamaiedge.net
Addresses: 2001:f40:a:c84::356e
           2001:f40:a:c85::356e
           2001:f40:a:c8e::356e
           2001:f40:a:c82::356e
           2001:f40:a:c81::356e
           211.25.121.218
Aliases:  www.microsoft.com
          www.microsoft.com-c-3.edgekey.net
          www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net
```

2. Run nslookup to determine the non-authoritative DNS servers for domain microsoft.com. Add screenshot to your answer.



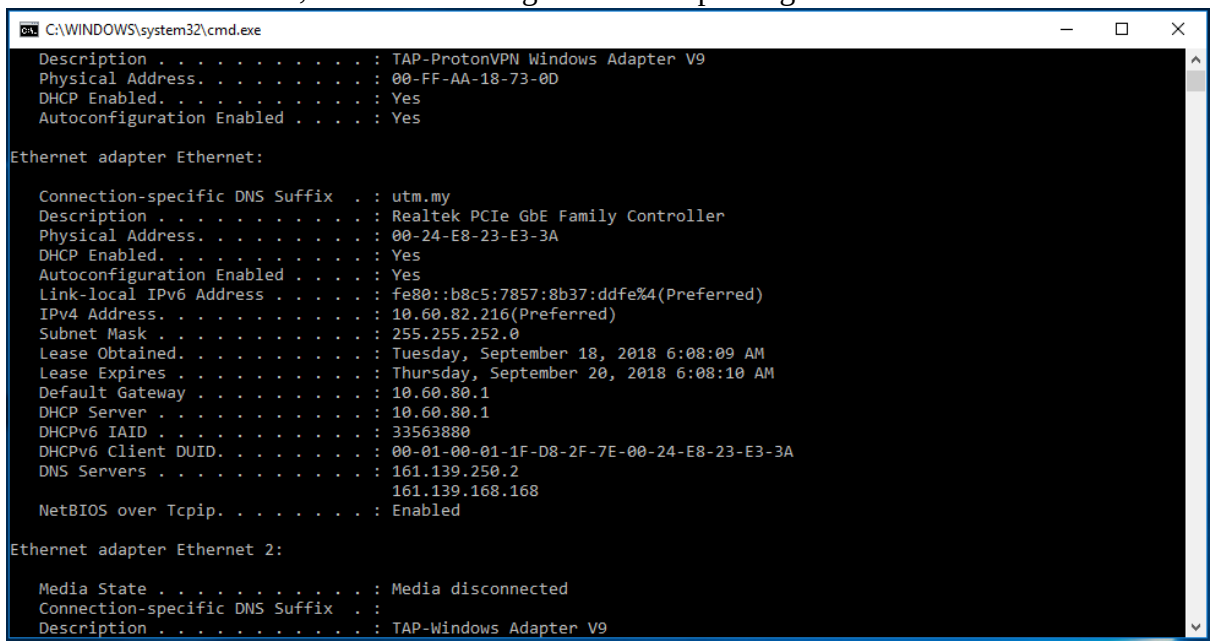
```
C:\Users\дания>nslookup -type=ns microsoft.com
Server:  ns1.utm.my
Address:  161.139.250.2

Non-authoritative answer:
microsoft.com    nameserver = ns1-39.azure-dns.com
microsoft.com    nameserver = ns2-39.azure-dns.net
microsoft.com    nameserver = ns3-39.azure-dns.org
microsoft.com    nameserver = ns4-39.azure-dns.info
```

2.0 ipconfig

ipconfig can be used to show your current TCP/IP information, including your address, DNS server addresses, adapter type and so on.

- Information about host, use the following command: `ipconfig /all`



```
C:\WINDOWS\system32\cmd.exe
Description . . . . . : TAP-ProtonVPN Windows Adapter V9
Physical Address. . . . . : 00-FF-AA-18-73-0D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : utm.my
    Description . . . . . : Realtek PCIe GbE Family Controller
    Physical Address. . . . . : 00-24-E8-23-E3-3A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::b8c5:7857:8b37:ddfe%4(Preferred)
    IPv4 Address. . . . . : 10.60.82.216(Preferred)
    Subnet Mask . . . . . : 255.255.252.0
    Lease Obtained. . . . . : Tuesday, September 18, 2018 6:08:09 AM
    Lease Expires . . . . . : Thursday, September 20, 2018 6:08:10 AM
    Default Gateway . . . . . : 10.60.80.1
    DHCP Server . . . . . : 10.60.80.1
    DHCPv6 IAID . . . . . : 33563880
    DHCPv6 Client DUID. . . . . : 00-01-00-01-1F-D8-2F-7E-00-24-E8-23-E3-3A
    DNS Servers . . . . . : 161.139.250.2
                           161.139.168.168
    NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
    Description . . . . . : TAP-Windows Adapter V9
```

Figure C.3: ipconfig /all result

- ipconfig is also very useful for managing the DNS information stored in your host. Each entry shows the remaining Time to Live (TTL) in seconds.
Command: `ipconfig /displaydns`

```

C:\WINDOWS\system32\cmd.exe
Record Name . . . . . : wpad.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 5917
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.163.188

-----
aimsweb.utm.my
Record Name . . . . . : aimsweb.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16483
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.18.215

-----
my.utm.my
Record Name . . . . . : my.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16484
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.20.240

C:\Users\pc-vastro220>

```

Figure C.4: ipconfig /displaydns result

- Flushing the DNS cache clears all entries and reloads the entries from the hosts file.

Command: ipconfig /flushdns

```

C:\WINDOWS\system32\cmd.exe
C:\Users\pc-vastro220>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\pc-vastro220>

```

Figure C.5: ipconfig /flushdns result

3.0 Tracing DNS with Wireshark

- Open packet trace file dns-trace-1. Answer the following questions.
1. Locate the DNS query and response messages. Are then sent over UDP or TCP? Add screenshots in your answer.

UDP

```

▶ Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
▶ Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
▶ Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23
▶ User Datagram Protocol, Src Port: 3163, Dst Port: 53
▶ Domain Name System (query)

```

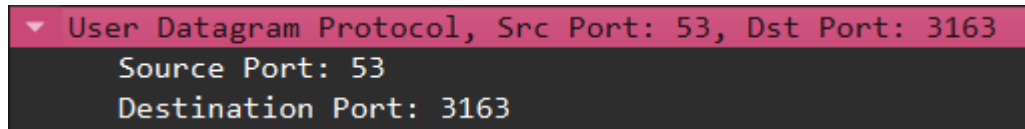
```

Frame 9: 104 bytes on wire (832 bits), 104 bytes captured (832 bits)
Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)
Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160
User Datagram Protocol, Src Port: 53, Dst Port: 3163
Domain Name System (response)

```

2. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

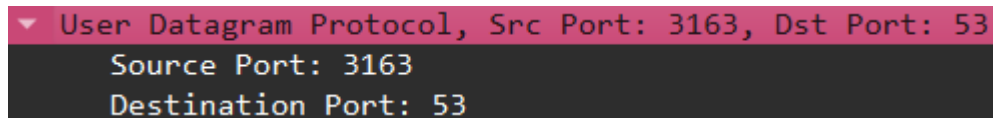
Packet No 8:



▼ User Datagram Protocol, Src Port: 53, Dst Port: 3163
Source Port: 53
Destination Port: 3163

This screenshot shows the details of a User Datagram Protocol (UDP) packet. The source port is 53 and the destination port is 3163.

Packet No 9:



▼ User Datagram Protocol, Src Port: 3163, Dst Port: 53
Source Port: 3163
Destination Port: 53

This screenshot shows the details of a User Datagram Protocol (UDP) packet. The source port is 3163 and the destination port is 53.

3. To what IP address is the DNS query message sent? Add screenshots in your answer.

No.	Time	Source	Destination
← 9	3.076689	128.238.29.23	128.238.38.160
→ 8	3.075845	128.238.38.160	128.238.29.23

128.238.38.160 and 128.238.29.23

4. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.

Standard Query, there are answers for packet 9.

Packet no 8:

```

Domain Name System (query)
  Transaction ID: 0x006e
  ▶ Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  ▶ Queries
    [Response In: 9]

```

Packet no 9:

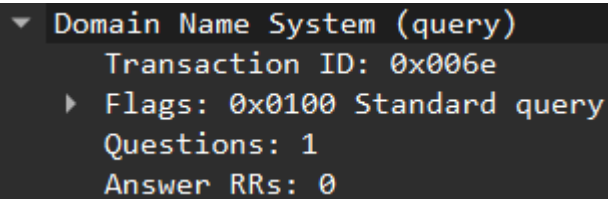
```

Domain Name System (response)
  Transaction ID: 0x006e
  ▶ Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 2
    Authority RRs: 0
    Additional RRs: 0
  ▶ Queries
  ▼ Answers
    ▶ www.ietf.org: type A, class IN, addr 132.151.6.75
    ▶ www.ietf.org: type A, class IN, addr 65.246.255.51
    [Request In: 8]
    [Time: 0.000844000 seconds]

```

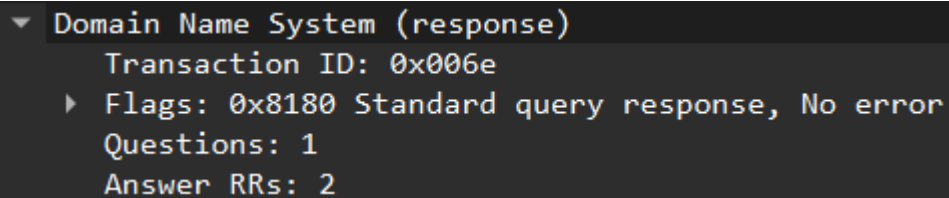
5. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

Packet no 8: 0

A screenshot of the Wireshark interface showing the details of packet 8. The packet is a Domain Name System (query). The transaction ID is 0x006e. The flags are 0x0100, indicating a standard query. There is 1 question and 0 answer RRs.

```
▼ Domain Name System (query)
  Transaction ID: 0x006e
  ► Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
```

Packet no 9: 2

A screenshot of the Wireshark interface showing the details of packet 9. The packet is a Domain Name System (response). The transaction ID is 0x006e. The flags are 0x8180, indicating a standard query response with no error. There is 1 question and 2 answer RRs.

```
▼ Domain Name System (response)
  Transaction ID: 0x006e
  ► Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 2
```

6. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message? Add screenshots in your answer.

No.	Time	Source	Destination	Protocol	Length	Info
27	3.191726	128.238.38.160	132.151.6.75	TCP	54	3370 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
26	3.191649	132.151.6.75	128.238.38.160	TCP	62	80 → 3370 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM

7. This web page contains images. Before retrieving each image, does your host issue new DNS queries?

```

▼ Hypertext Transfer Protocol
  ▶ GET / HTTP/1.1\r\n
    Accept: image/gif, image/x-bitmap, image/jpeg, image/pjpeg, application/vnd.ms-excel,
    Accept-Language: en-us\r\n
    Accept-Encoding: gzip, deflate\r\n
    User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; .NET CLR 1.0.3705)\r\n
    Host: www.ietf.org\r\n

```

- Open packet trace file dns-trace-2 for nslookup.
 - We see from Wireshark that nslookup actually sent three DNS queries and received three DNS responses. For the purpose of this lab, ignore the first two sets of queries/responses, as they are specific to nslookup and are not normally generated by standard Internet applications. You should instead focus on the last query and response messages.
 - Answer the following questions.
8. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

Destination port : 53

```

▼ User Datagram Protocol, Src Port: 3742, Dst Port: 53
  Source Port: 3742
  Destination Port: 53

```

Source port : 53

```

▼ User Datagram Protocol, Src Port: 53, Dst Port: 3742
  Source Port: 53
  Destination Port: 3742

```

9. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? Add screenshots in your answer.

IP address is the DNS query message sent : 128.238.29.22

```
Source Address: 128.238.38.160
Destination Address: 128.238.29.22
```

10. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.

Type of DNS query : standard query.

It does not contain any answer

```
► Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
```

11. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

Answer provided : 1

The answer contain name of the website which is www.mit.edu and the address of the website which is 18.7.22.83

```
► Flags: 0x8580 Standard query response, No error
Questions: 1
Answer RRs: 1
Authority RRs: 3
Additional RRs: 3
▼ Queries
  ► www.mit.edu: type A, class IN
▼ Answers
  ► www.mit.edu: type A, class IN, addr 18.7.22.83
```